

MEKORYUK, AK, USA

WMO: 702185

Lat: 60.372N

Lon: 166.272W

Elev: 14

StdP: 101.16

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26622

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.1	-23.0	-30.2	0.2	-23.8	-27.3	0.3	-22.9	18.3	-4.8	16.7	-4.0	4.0	30	0.962

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.8	16.1	12.7	14.1	11.4	13.0	10.8	13.3	15.4	12.2	13.9	11.3	13.0	5.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	8.9	14.4	11.2	8.3	13.5	10.0	7.7	12.5	37.2	15.4	34.5	14.2	32.2	13.0	19.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.1	14.5	12.7	-28.0	19.1	2.2	2.8	-29.6	21.1	-31.0	22.7	-32.2	24.3	-33.8	26.3		
			DB	WB	-28.2	14.8	2.0	1.7	-29.6	16.0	-30.8	17.0	-31.9	17.9	-33.3	19.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.6	-11.1	-9.1	-9.2	-3.9	2.0	6.5	9.4	10.3	7.7	1.6	-3.4	-8.3
	DBStd	8.92	7.33	6.70	6.42	4.85	3.01	2.55	1.94	2.05	2.39	3.41	4.71	6.56
	HDD10.0	3916	654	534	596	416	248	109	33	20	73	260	403	569
	HDD18.3	6904	913	767	854	666	507	356	276	249	318	518	653	827
	CDD10.0	53	0	0	0	0	0	3	16	29	6	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	6.3	7.0	7.1	6.4	6.3	5.2	4.9	4.9	5.8	6.3	6.7	7.4	7.1
Precipitation	PrecAvg	714	49	48	34	30	38	54	77	98	96	71	66	53
	PrecMax	974	110	117	107	60	82	82	117	147	170	142	130	111
	PrecMin	573	10	14	4	7	4	24	46	34	14	20	13	12
	PrecStd	94	24	26	22	13	18	15	21	26	29	25	30	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.3	2.1	2.7	4.5	12.7	17.3	17.6	18.0	13.9	9.3	7.2	3.2
		MCWB	0.8	1.0	1.1	2.6	8.5	11.9	12.8	14.6	10.8	8.3	5.6	2.1
	2%	DB	1.3	1.3	1.5	2.9	9.6	13.8	15.9	16.2	12.8	8.0	5.0	2.3
		MCWB	0.2	0.4	0.4	1.1	7.0	10.0	12.3	13.3	10.5	6.7	3.9	1.3
	5%	DB	1.0	1.0	0.9	2.3	7.7	12.3	13.9	14.7	12.2	7.4	3.7	1.3
		MCWB	-0.1	0.2	0.0	0.8	5.5	9.3	11.3	12.4	10.2	5.8	2.5	0.4
	10%	DB	-0.9	0.1	-0.3	1.4	6.4	11.0	12.7	13.3	11.3	6.4	2.5	0.8
		MCWB	-1.4	-0.5	-0.7	0.2	4.3	8.5	10.7	11.4	9.5	4.3	1.4	-0.1
	0.4%	WB	0.8	1.2	1.5	3.2	8.9	12.4	13.9	15.1	12.1	8.7	5.7	2.6
		MCDB	1.4	1.4	2.2	4.6	12.3	16.2	16.3	17.4	12.9	9.2	6.9	3.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	0.2	0.5	0.6	1.5	7.0	10.5	12.7	13.9	10.9	7.5	3.9	1.1
		MCDB	1.0	0.9	1.3	2.7	9.3	13.2	15.1	16.0	12.3	8.0	5.1	1.8
	5%	WB	-0.2	0.1	-0.1	0.8	5.7	9.4	11.7	12.8	10.3	6.0	2.9	0.4
		MCDB	0.7	0.7	0.6	1.8	7.7	11.9	13.6	14.5	11.7	7.3	3.7	1.2
	10%	WB	-1.4	-0.6	-1.0	0.3	4.7	8.4	10.9	11.8	9.6	4.8	1.4	-0.2
		MCDB	-0.9	0.0	-0.5	1.3	6.6	10.4	12.6	13.3	11.0	6.6	2.4	0.6
Mean Daily Temperature Range	MDBR		5.7	5.7	6.0	5.1	4.9	6.0	5.3	4.8	4.5	4.4	4.7	5.4
	5% DB	MCDBR	5.0	5.0	4.4	4.5	7.7	9.5	7.7	6.9	5.3	5.4	4.6	4.5
		MCWBR	4.8	4.9	4.1	3.8	5.7	6.6	5.4	5.0	4.0	4.6	4.3	4.5
	5% WB	MCDBR	5.2	5.3	4.3	4.2	7.6	8.8	6.9	6.6	4.8	5.1	4.5	4.6
		MCWBR	5.1	5.2	4.1	3.6	5.7	6.3	5.0	4.9	3.9	4.7	4.4	4.5
Clear-Sky Solar Irradiance	taub		0.236	0.292	0.294	0.360	0.374	0.377	0.393	0.396	0.350	0.315	0.281	0.237
	taud		2.068	2.110	2.186	2.045	2.228	2.381	2.366	2.332	2.424	2.333	2.066	1.928
	Ebn at Noon		521	682	831	824	847	853	827	790	770	666	430	321
	Edh at Noon		43	76	97	136	123	108	108	103	79	62	43	32
All-Sky Solar Radiation	RadAvg		0.36	1.07	2.72	4.23	4.77	5.04	4.11	3.20	2.11	1.09	0.35	0.16
	RadStd		0.06	0.14	0.33	0.44	0.53	0.44	0.34	0.41	0.22	0.06	0.05	0.02

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page